

Control and amplification of Bloch oscillations via photon-mediated interactions: Supplemental Materials

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S1. CAVITY QED WITH WANNIER-STARK STATE

A. Dispersive coupling between atoms and cavity

In the main text, we considered N ultracold atoms trapped in a standing-wave optical cavity along the vertical direction $\hat{z}$. The atoms are assumed to be confined in the ground band of the one-dimensional lattice with lattice depth V_0 and wave vector $k_l = 2\pi/\lambda_l$. A single internal level $|g\rangle$ in the atomic ground manifold is coupled to an atomic excited state $|e\rangle$ with a transition energy $\hbar\omega_0 = \hbar(\omega_e - \omega_g)$, via a single cavity mode $\hat{a}$ with angular frequency ω_c and wavelength λ_c . The atom-cavity coupling has spatial dependence $\mathcal{G}(z) = \mathcal{G}_0 \sin(k_c z)$, with $k_c = 2\pi/\lambda_c$. The cavity mode is coherently driven by an external light field with detuning $\Delta_c = \omega_p - \omega_c$ from the bare cavity mode, which generates a net injected field in the cavity with amplitude η_p . The full atom-cavity Hamiltonian is given as $H = \hat{H}_{\text{atom}} + \hat{H}_{\text{light}} + \hat{H}_{\text{int}}$. each of the terms can be written as :

$$\hat{H}_{\text{light}} = \hbar (\eta_p \hat{a}^\dagger e^{-i\omega_p t} + \eta_p^* \hat{a} e^{i\omega_p t}) + \hbar\omega_c \hat{a}^\dagger \hat{a} \quad (\text{S1})$$

$$\hat{H}_{\text{atom}} = \sum_{\tau=g,e} \int dz \hat{\psi}_\tau^\dagger(z) \left[\frac{p^2}{2M} + V(z) + \hbar\omega_\tau \right] \hat{\psi}_\tau(z) \quad (\text{S2})$$

$$\hat{H}_{\text{int}} = \hbar \int dz \mathcal{G}_0 \sin k_c z \left[\hat{a} \hat{\psi}_e^\dagger(z) \hat{\psi}_g(z) + \hat{a}^\dagger \hat{\psi}_g^\dagger(z) \hat{\psi}_e(z) \right]. \quad (\text{S3})$$

Here $V(z) = Mgz + V_0 \sin^2 k_l z$ describes the external potentials experienced by the atoms. $\hat{\psi}_{e(g)}^\dagger(z)$ is the field operator that creates an atom in the state $e(g)$ at position z , $\omega_{e(g)}$. Under the rotating frame of the pump field (set by the Hamiltonian $H_0 = \hbar\omega_p \hat{a}^\dagger \hat{a} + \hbar\omega_p \int dz \hat{\psi}_e^\dagger(z) \hat{\psi}_e(z)$), the system's Hamiltonian takes the following form:

$$\hat{H} = \hbar (\eta_p \hat{a}^\dagger + \eta_p^* \hat{a}) - \Delta_c \hbar \hat{a}^\dagger \hat{a} - \hbar\Delta_0 \int dz \hat{\psi}_e^\dagger(z) \hat{\psi}_e(z) + \sum_{\tau=g,e} \int dz \hat{\psi}_\tau^\dagger(z) \left[\frac{p^2}{2M} + V(z) \right] \hat{\psi}_\tau(z) \quad (\text{S4})$$

$$+ \hbar \int dz \mathcal{G}_0 \sin k_c z \left[\hat{a} \hat{\psi}_e^\dagger(z) \hat{\psi}_g(z) + \hat{a}^\dagger \hat{\psi}_g^\dagger(z) \hat{\psi}_e(z) \right], \quad (\text{S5})$$

where we defined the detuning of the pump from the atomic transition as $\Delta_0 = \omega_p - \omega_0$.

Furthermore, under the assumption $\Delta_0 \gg \mathcal{G}_0 \sqrt{\langle \hat{a}^\dagger \hat{a} \rangle}$ and $\Delta_0 \gg \gamma$ with γ the excited state spontaneous emission rate, the excited state population remains negligible during the relevant time scales. In this limit we can adiabatic eliminate the excited state $|e\rangle$ ($\hat{\psi}_e(z) \approx \mathcal{G}_0 \hat{a} \hat{\psi}_g(z) \sin k_c z / \Delta_0$), which leads to the following effective Hamiltonian acting on the ground state $|g\rangle$ manifold only,

$$\hat{H} = -\hbar\Delta_c \hat{a}^\dagger \hat{a} + \hbar (\eta_p \hat{a}^\dagger + \eta_p^* \hat{a}) + \int dz \hat{\psi}_g^\dagger(z) \left[\frac{\hbar(\mathcal{G}_0 \sin k_c z)^2}{\Delta_0} \hat{a}^\dagger \hat{a} + \frac{p^2}{2M} + V(z) \right] \hat{\psi}_g(z). \quad (\text{S6})$$

In the tight-binding limit, the resulting single-particle eigenstates of the Hamiltonian $p^2/2m + V(z)$ become the so-called Wannier-Stark (WS) states $|\phi_n\rangle$ ($n \in \mathbb{Z}$):

$$E_n = Mga_l n, \quad \phi_n(z) = \sum_m \mathcal{J}_{m-n} \left(\frac{2J_0}{Mga_l} \right) w(z - ma_l). \quad (\text{S7})$$

Here $\mathcal{J}_n$ denotes the Bessel function of the first kind, J_0 denotes the nearest-neighbor couplings in the ground band, $a_l = \lambda_l/2$ is the lattice spacing and $w(z)$ is the ground band Wannier function. We will also use $E_R = (\hbar k_l)^2/2M$ for the atomic recoil energy.

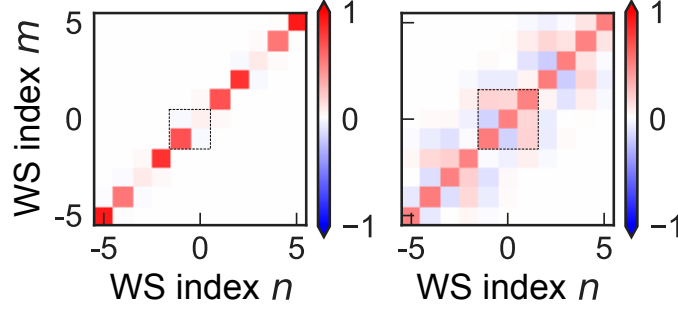


FIG. S1. The coupling coefficient $J_{m,n}$ for ^{87}Rb atoms ($\lambda_l = 532$ nm, $\lambda_c = 780$ nm). Left: $V_0 = 20E_R$ and right: $V_0 = 6E_R$. Start from $|\phi_0\rangle$, the many-body dynamics mainly happens within the dashed square for either two-level model (left, deep lattice region for dynamical phase transitions) and three-level model (right, shallow lattice region for amplification of Bloch oscillations).

The field operator, when written in the WS basis takes the form, $\hat{\psi}_g(z) = \sum_n \hat{c}_n \phi_n(z)$, where $\hat{c}_n$ annihilates an atom in the state $|\phi_n\rangle$. In this basis we can rewrite the Hamiltonian [Eq. (S6)] as:

$$\hat{H}/\hbar = -\Delta_c \hat{a}^\dagger \hat{a} + \eta_p \hat{a}^\dagger + \eta_p^* \hat{a} + \frac{\mathcal{G}_0^2}{\Delta_0} \hat{a}^\dagger \hat{a} \sum_{m,n} J_{m,n} \hat{c}_m^\dagger \hat{c}_n + \omega_B \sum_n n \hat{c}_n^\dagger \hat{c}_n, \quad (\text{S8})$$

where $J_{m,n} = \int dz \phi_m(z) \phi_n(z) \sin^2 k_c z$ describes the overlap between the WS states $|\phi_m\rangle, |\phi_n\rangle$ weighted by the cavity field mode function. In Fig. S1 we show the value of these couplings for the typical lattice depths we work in this paper. We define the effective particle number:

$$\hat{N}_{\text{eff}} = \sum_{m,n} J_{m,n} \hat{c}_m^\dagger \hat{c}_n, \quad (\text{S9})$$

as the effective number of atoms coupled to the cavity, which shifts the cavity resonance frequency by $\mathcal{G}_0^2 \langle \hat{N}_{\text{eff}} \rangle / \Delta_0$.

B. Adiabatic elimination of cavity field

Here we study the dynamics via Heisenberg equations of motion using a Markovian approximation. We adiabatic eliminate the cavity field using the fact that Δ_c sets the largest frequency scale and derive the effective atom-only Hamiltonian. To do that, we formally integrate the Heisenberg equation of motion of the cavity mode operator $\hat{a}$ and photon number operators $\hat{a}^\dagger \hat{a}$, then plug them back into the Hamiltonian [Eq. (S8)]. We remove the fast rotating terms which relax much faster than the time it takes an atom to perform a BO.

The Heisenberg-Langevin equation of the motion for the cavity mode $\hat{a}$ is given by:

$$\frac{d}{dt} \hat{a} = i[\hat{H}/\hbar, \hat{a}] + \left(\frac{\kappa}{2} \hat{a}^\dagger + \hat{f}^\dagger\right)[\hat{a}, \hat{a}] - [\hat{a}, \hat{a}^\dagger] \left(\frac{\kappa}{2} \hat{a} + \hat{f}\right) = i(\Delta_c - \frac{\mathcal{G}_0^2 \hat{N}_{\text{eff}}}{\Delta_0}) \hat{a} - i\eta_p - \frac{\kappa}{2} \hat{a} + \hat{f}, \quad (\text{S10})$$

with κ for the cavity decay rate. The above equation captures the dissipative dynamics generated by κ along with the quantum Langevin noise operator $\hat{f}$, which gives the formal solution for the cavity field operator:

$$\begin{aligned} \hat{a} &= -i\eta_p \exp \left[i \int_0^t d\tau \left(\hat{\Delta} + i\kappa/2 \right) \right] \int_0^t dt' \exp \left[-i \int_0^{t'} d\tau \left(\hat{\Delta} + i\kappa/2 \right) \right] + \hat{f}' \\ &\approx \frac{\eta_p}{\hat{\Delta} + i\kappa/2} + \hat{f}' \end{aligned} \quad (\text{S11})$$

with

$$\hat{\Delta} = \Delta_c - \frac{\mathcal{G}_0^2 \hat{N}_{\text{eff}}}{\Delta_0}, \quad (\text{S12})$$

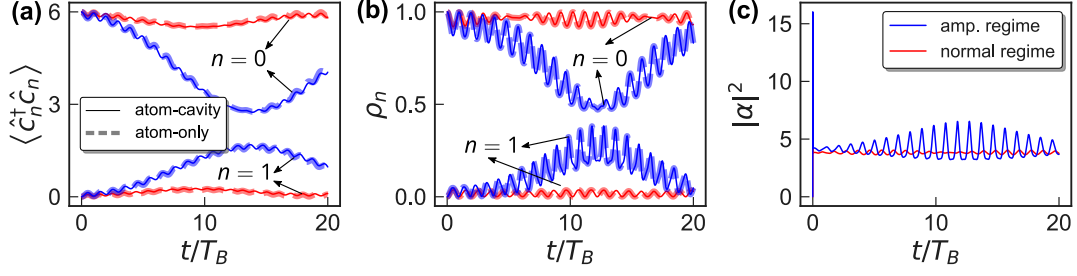


FIG. S2. Benchmarks of the atom-cavity Hamiltonian [Eq. (S8)] with the effective atom-only Hamiltonian [Eq. (S14)]. The red curves for $V_0 = 5.8E_R$ (normal regime) and the blue curves for $V_0 = 6.2E_R$ (amplification regime) are used for all the simulations in the figure. (a) Exact Diagonalization (ED) simulation of the dynamics for 6 particles in 3 WS states with the initial state $(\hat{c}_0^\dagger)^6 |\text{vac}\rangle$. Populations $\langle \hat{c}_0^\dagger \hat{c}_0 \rangle$ (start from 6) as well as $\langle \hat{c}_1^\dagger \hat{c}_1 \rangle$ (start from 0) for these two lattice depths are plotted. The solid lines are the exact simulations under the Hamiltonian Eq. (S8) (with the photon space $n_{\text{cut}} = 10$) and the dashed lines under the Hamiltonian Eq. (S14). (b) Mean-field dynamics of ρ_n with initial state $|\phi_0\rangle$. The solid lines are simulated with the atom-cavity mean-field equations of motion [Eq. (S18)] and the dashed lines are simulated with atom-only equations of motion [Eq. (S17)]. Populations ρ_0 (start from 1) as well as ρ_1 (start from 0) for these two parameters are plotted. The differences between the atom-cavity and atom-only simulations can be ignored for both (a) and (b). (c) Mean-field evolution for the cavity photon number with the same parameters for the red and blue curves as in (b).

Here $\hat{f}'$ is another quantum Langevin noise operator. Below we consider the regime $\Delta_c, \kappa \gg \omega_B$ where the cavity-field dynamics evolve much faster than the atomic dynamics, thus it follows the latter adiabatically. As a result, we can obtain the formal solution for the cavity photon number operator:

$$\hat{a}^\dagger \hat{a} = \frac{|\eta_p|^2}{\hat{\Delta}^2 + (\kappa/2)^2} + \hat{g} \approx \frac{|\eta_p|^2}{\hat{\Delta}^2}. \quad (\text{S13})$$

One more time $\hat{g}$ represents a different quantum Langevin noise operator. For the last approximation above, we focus of the regime $\Delta_c \gg \kappa$ where the unitary dynamics dominates and we can ignore the dissipation process to leading order.

If we insert the above solution of the cavity field into Eq. (S8), the effective atom-only Hamiltonian in the Schrodinger picture can be written as:

$$\begin{aligned} \hat{H}_{\text{eff}}/\hbar &= \omega_B \sum_n n \hat{c}_n^\dagger \hat{c}_n + \eta_p \frac{\eta_p^*}{\hat{\Delta}} + \eta_p^* \frac{\eta_p}{\hat{\Delta}} + \left(\mathcal{G}_0^2 \hat{N}_{\text{eff}}/\Delta_0 - \Delta_c \right) \frac{|\eta_p|^2}{\hat{\Delta}^2} \\ &= \omega_B \sum_n n \hat{c}_n^\dagger \hat{c}_n + 2 \frac{|\eta_p|^2}{\hat{\Delta}} - \hat{\Delta} \frac{|\eta_p|^2}{\hat{\Delta}^2} \\ &= \omega_B \sum_n n \hat{c}_n^\dagger \hat{c}_n + \frac{|\eta_p|^2}{\Delta_c - \mathcal{G}_0^2 \hat{N}_{\text{eff}}/\Delta_0} \\ &\equiv \omega_B \sum_n n \hat{c}_n^\dagger \hat{c}_n + \hat{V}_{\text{cav}}(\hat{N}_{\text{eff}}) \end{aligned} \quad (\text{S14})$$

where $\hat{V}_{\text{cav}}(\hat{N}_{\text{eff}}) = -(VN/\beta)/(1 + \beta \hat{N}_{\text{eff}}/N)$ is the dynamical potential induced by the cavity which depends on the atomic motion. $\hat{V}_{\text{cav}}$ is parameterized by the maximum AC Stark shift introduced by the bare cavity mode, $V = \mathcal{G}_0^2 |\eta_p|^2 / (\Delta_c^2 \Delta_0)$, as well as by the ratio between the maximum cavity shift and the bare cavity detuning, $\beta = -N \mathcal{G}_0^2 / (\Delta_0 \Delta_c)$. We assume $\beta > 0$ (Δ_0 and Δ_c have opposite signs) to avoid hitting a resonance.

As a benchmark for the effective atom-only Hamiltonian derived in Eq. (S14), we compare the exact dynamics for 6 particles in 3 WS states under Eq. (S8) and Eq. (S14) in Fig. S2(a). In the simulation, we choose $\Delta_c = 400\omega_B$, $\kappa = 20\Delta_c$, $\eta_p = \Delta_c/10$ as well as $\mathcal{G}_0^2/\Delta_0 = -100\omega_B$ in the atom-cavity simulation (dashed lines), which corresponds to $V = \mathcal{G}_0^2 |\eta_p|^2 / (\Delta_c^2 \Delta_0) = \omega_B$ and $\beta = -N \mathcal{G}_0^2 / (\Delta_0 \Delta_c) = 1.5$ in the atom-only simulation (solid lines). The simulation results match well with each other for the lattice depth in the normal regime (red curves) and amplification regime (blue curves), which verify the effectiveness of the atom-only Hamiltonian.

C. Equations of motion for atoms

To simulate the dynamics under Eq. (S14), we can calculate the equations of motion for the field operators $\hat{c}_m$ as:

$$\begin{aligned} i\dot{\hat{c}}_m &= m\omega_B \hat{c}_m - \frac{VN}{\beta} \left[\hat{c}_m, \frac{1}{(1 + \beta \hat{N}_{\text{eff}}/N)} \right] \\ &= m\omega_B \hat{c}_m - \frac{VN}{\beta} \left[\hat{c}_m, 1 - \frac{\beta}{N} \hat{N}_{\text{eff}} + \left(\frac{\beta}{N}\right)^2 \hat{N}_{\text{eff}}^2 - \left(\frac{\beta}{N}\right)^3 \hat{N}_{\text{eff}}^3 + \dots \right], \end{aligned} \quad (\text{S15})$$

Then we can simplify the equations above with $[\hat{c}_m, \sum_{p,q} J_{p,q} \hat{c}_p^\dagger \hat{c}_q] = \sum_n J_{m,n} \hat{c}_n$:

$$\begin{aligned} i\dot{\hat{c}}_m &= m\omega_B \hat{c}_m - \frac{VN}{\beta} \left\{ -\frac{\beta}{N} [\hat{c}_m, \hat{N}_{\text{eff}}] + \left(\frac{\beta}{N}\right)^2 [\hat{c}_m, \hat{N}_{\text{eff}}^2] - \left(\frac{\beta}{N}\right)^3 [\hat{c}_m, \hat{N}_{\text{eff}}^3] + \dots \right\} \\ &= m\omega_B \hat{c}_m - \frac{VN}{\beta} \left\{ -\frac{\beta}{N} \sum_n J_{mn} \hat{c}_n + \left(\frac{\beta}{N}\right)^2 \left(\sum_n J_{mn} \hat{c}_n \hat{N}_{\text{eff}} + \hat{N}_{\text{eff}} \sum_n J_{mn} \hat{c}_n \right) + \dots \right\}. \end{aligned} \quad (\text{S16})$$

Finally, we apply the mean-field approximation to the operators $\langle \sum_n J_{mn} \hat{c}_n \hat{N}_{\text{eff}} \rangle \approx \langle \sum_n J_{mn} \hat{c}_n \rangle \langle \hat{N}_{\text{eff}} \rangle$, and obtain the following equations of motion:

$$i\langle \dot{\hat{c}}_m \rangle = m\omega_B \langle \hat{c}_m \rangle + \frac{V}{(1 + \beta \langle \hat{N}_{\text{eff}} \rangle / N)^2} \sum_{m,n} J_{m,n} \langle \hat{c}_n \rangle, \quad (\text{S17})$$

All the results in the main text were obtained by solving the mean-field equations of motion written above.

Meanwhile, the mean-field equations for the atom-cavity Hamiltonian [Eq. (S8)] is given by,

$$\begin{aligned} i\dot{\alpha} &= - \left(\Delta_c + i\frac{\kappa}{2} - \frac{\mathcal{G}_0^2 \langle \hat{N}_{\text{eff}} \rangle}{\Delta_0} \right) \alpha + \eta_p \\ i\langle \dot{\hat{c}}_m \rangle &= m\omega_B \langle \hat{c}_m \rangle + \frac{\mathcal{G}_0^2}{\Delta_0} |\alpha|^2 \sum_{m,n} J_{m,n} \langle \hat{c}_n \rangle, \end{aligned} \quad (\text{S18})$$

with $\alpha = \langle \hat{a} \rangle$. We compare the mean-field dynamics Eq. (S17) and Eq. (S18) in Fig. S2(b). In the simulation, we choose reasonable experimental parameters $N = 2 \times 10^4$ atoms, $\Delta_c = 2\pi \times 2$ MHz, $\kappa = \Delta_c/20$, $\eta_p = 3\Delta_c$ as well as $\mathcal{G}_0^2/\Delta_0 = -2\pi \times 100$ Hz in the atom-cavity simulation [Eq. (S18)], which corresponds to $V = \mathcal{G}_0^2 |\eta_p|^2 / (\Delta_c^2 \Delta_0) = 1.57\omega_B$ and $\beta = -N\mathcal{G}_0^2 / (\Delta_0 \Delta_c) = 1$ in the atom-only simulation [Eq. (S17)]. Still, the simulations for both normal regime and amplification regime match with each other pretty well with the difference can be ignored, which again verifies the validation of the effective atom-only Hamiltonian. In Fig. S2(c), we plot the evolution of cavity photon number $|\alpha|^2$ which follows the atomic motion adiabatically.

S2. DYNAMICAL PHASE TRANSITION WITH WANNIER-STARK STATES

In this part, we consider the deep lattice regime and discuss how to map the atom-only Hamiltonian to a spin model. As discussed in the main text, for a deep lattice $V_0 = 20E_R$, the WS states approach the Wannier orbitals which are localized. The overlap integral $J_{m,n}$ for $V_0 = 20E_R$ is shown in the left panel of Fig. S1 with $J_{0,0} \approx 0 \ll J_{1,1} \approx J_{-1,-1}$ and $J_{1,0} \approx -J_{0,-1} \approx 0$. As a result non-trivial dynamics happens only for $V(J_{n,n} - J_{0,0}) + n\omega_B \approx 0$ when starting from $|\phi_0\rangle$. Here we consider $V > 0$ and deal with two bosonic modes $\hat{c}_0, \hat{c}_{-1}$. For simplicity, we define $\Omega_n = J_{n,n+1}$ as well as $\Delta_n = (J_{n,n} - J_{0,0})/2$. The spin operators are defined as follows,

$$\begin{aligned} \hat{S}_x &= \frac{1}{2}(\hat{c}_{-1}^\dagger \hat{c}_0 + \hat{c}_0^\dagger \hat{c}_{-1}) \\ \hat{S}_y &= -\frac{i}{2}(\hat{c}_{-1}^\dagger \hat{c}_0 - \hat{c}_0^\dagger \hat{c}_{-1}) \\ \hat{S}_z &= \frac{1}{2}(\hat{c}_{-1}^\dagger \hat{c}_{-1} - \hat{c}_0^\dagger \hat{c}_0), \end{aligned} \quad (\text{S19})$$

and the total particle number $\hat{N} = \hat{c}_{-1}^\dagger \hat{c}_{-1} + \hat{c}_0^\dagger \hat{c}_0$. Such pseudospin operators satisfy the SU(2) algebra and we can rewrite the effective number operator as ($\bar{\omega} = (J_{-1,-1} + J_{0,0})/2$):

$$\hat{N}_{\text{eff}} = 2\Omega_{-1}\hat{S}_x + 2\Delta_{-1}\hat{S}_z + \bar{\omega}N, \quad (\text{S20})$$

as well as the effective spin model from Eq. (S8) in terms only of $\hat{c}_{-1}$ and $\hat{c}_0$:

$$\hat{H}_{\text{eff}}/\hbar = -\omega_B\hat{S}_z + \hat{V}_{\text{cav}}(\hat{N}_{\text{eff}}) \quad (\text{S21})$$

Similar in Eq. (S15), we can derive Heisenberg equations of motion for the collective spin operator $\hat{S}_{x,y,z}$. Using the mean-field approximation which neglects the quantum correlation between different spins we obtain,

$$\begin{aligned} \langle \dot{\hat{S}}_x \rangle &= (\omega_B - 2\Delta_{-1}\tilde{V}) \langle \hat{S}_y \rangle \\ \langle \dot{\hat{S}}_y \rangle &= (2\Delta_{-1}\tilde{V} - \omega_B) \langle \hat{S}_x \rangle - 2\Omega_{-1}\tilde{V} \langle \hat{S}_z \rangle \\ \langle \dot{\hat{S}}_z \rangle &= 2\Omega_{-1}\tilde{V} \langle \hat{S}_y \rangle. \end{aligned} \quad (\text{S22})$$

Moreover, if we introduce the mean-field real variable $s_\alpha = 2 \langle \hat{S}_\alpha \rangle / N$, $\alpha \in \{x, y, z\}$, the above equations become:

$$\begin{aligned} \dot{s}_x &= (\omega_B - 2\Delta_{-1}\tilde{V})s_y \\ \dot{s}_y &= (2\Delta_{-1}\tilde{V} - \omega_B)s_x - 2\Omega_{-1}\tilde{V}s_z \\ \dot{s}_z &= 2\Omega_{-1}\tilde{V}s_y, \end{aligned} \quad (\text{S23})$$

with

$$\tilde{V} = \frac{V}{(1 + \beta N_{\text{eff}}(t)/N)^2}, \quad (\text{S24})$$

here $N_{\text{eff}}(t) = N(\Omega_{-1}s_x + \Delta_{-1}s_z + \bar{\omega})$. Later we will use the symbol $n_{\text{eff}} \equiv \Omega_{-1}s_x + \Delta_{-1}s_z + \bar{\omega}$ for convenience. We compare the results from Eq. (S17) and Eq. (S23) to numerical simulations of the full Hamiltonian and they match with each other, which means the two-mode approximation works in this case. Now, we discuss the dynamical phase transition predicted by Eq. (S23). Using both energy conservation as well as the identity $(\hat{S}^x)^2 + (\hat{S}^y)^2 + (\hat{S}^z)^2 = (\frac{N}{2} + 1)\frac{N}{2}$ in the large N limit, the real variable (s_x, s_y, s_z) with initial condition $s_z = -1, s_x = s_y = 0$ satisfy the following two conservation laws:

$$s_x^2 + s_y^2 + s_z^2 = 1 \quad (\text{S25})$$

$$-\omega_B s_z - \frac{2V/\beta}{1 + \beta n_{\text{eff}}} = \omega_B - \frac{2V/\beta}{1 + \beta(\bar{\omega} - \Delta_{-1})}, \quad (\text{S26})$$

then we can express $s_{x,y,z}$ all as a function of n_{eff} :

$$\begin{aligned} s_z(n_{\text{eff}}) &= \frac{F(\bar{\omega} - \Delta_{-1}) - F(n_{\text{eff}})}{\omega_B} - 1 \\ s_x(n_{\text{eff}}) &= \frac{n_{\text{eff}} - \Delta_{-1}s_z(n_{\text{eff}}) - \bar{\omega}}{\Omega_{-1}} \\ s_y^2(n_{\text{eff}}) &= 1 - s_x^2(n_{\text{eff}}) - s_z^2(n_{\text{eff}}), \end{aligned} \quad (\text{S27})$$

also we define a function,

$$F(x) = \frac{2V/\beta}{1 + \beta x}. \quad (\text{S28})$$

The dynamics correspond to a classical particle moving in the external potential from Eq. (S23):

$$(n_{\text{eff}})^2 + f(n_{\text{eff}}) = 0, \quad (\text{S29})$$

with the potential $f(n_{\text{eff}}) = -(\omega_B\Omega_{-1})^2 s_y^2(n_{\text{eff}})$. The condition $f(n_{\text{eff}}) = 0$ determines the roots and we find $n_{\text{eff}}^0 = \bar{\omega} - \Delta_{-1}$ is one of such root. The effective potential can have either two or four solutions within the region $n_{\text{eff}} \in$

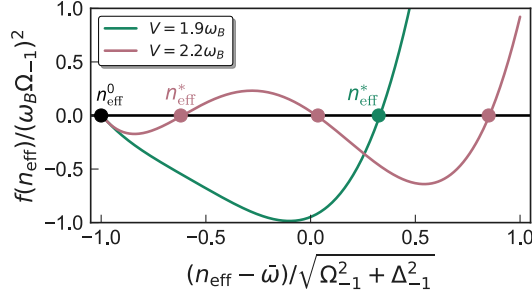


FIG. S3. Number of roots for the effective potential $f(n_{\text{eff}})$. In the case of $V = 1.9\omega_B$, $\beta = 0.5$ (green), $f(n_{\text{eff}})$ has two roots; In the case of $V = 2.1\omega_B$, $\beta = 0.5$ (red), $f(n_{\text{eff}})$ has four roots. The nearest root is labelled by n_{eff}^* and the jump of n_{eff}^* indicate DPTs.

$[\bar{\omega} - \sqrt{\Omega_{-1}^2 - \Delta_{-1}^2}, \bar{\omega} + \sqrt{\Omega_{-1}^2 + \Delta_{-1}^2}]$ shown in Fig. S3, and the dynamics of n_{eff} can be understood as the oscillations between n_{eff}^0 and the nearest root n_{eff}^* . Begin with a function $f(n_{\text{eff}})$ with two roots, and continuously tune the parameters of $f(n_{\text{eff}})$ so that two new roots appear in between, then a jump of the nearest root n_{eff}^* should occur during this process. The dynamical paramagnetic phase corresponds to four roots, while the dynamical ferromagnetic phase corresponds to two roots. Deep in the dynamical paramagnetic phase, one can access the whole Bloch sphere ($n_{\text{eff}}^* \approx \bar{\omega} + \sqrt{\Omega_{-1}^2 + \Delta_{-1}^2}$), while deep in the dynamical ferromagnetic phase, the Bloch vector only cycles around the south pole ($n_{\text{eff}}^* \approx n_{\text{eff}}^0$). By numerically varying β and V , we compute the number of roots to produce the phase diagram featured in the main text. If $\beta < 0.32$, $f(n_{\text{eff}})$ can only possess two roots, resulting in a smooth crossover rather than a phase transition.

The LMG model [S1, S2] supports a dynamical phase transition with Hamiltonian:

$$H_{\text{LMG}} = \chi \hat{S}_z^2 + \Omega \hat{S}_x - \delta \hat{S}_z. \quad (\text{S30})$$

To gain more insights on how our model related to the LMG model, we can expand Eq. (S8) to first order and second order in $(2\Omega_{-1}\hat{S}_x + 2\Delta_{-1}\hat{S}_z)$. To the first order:

$$\hat{H}_{\text{eff}} = -\omega_B \hat{S}_z + V(2\Omega_{-1}\hat{S}_x + 2\Delta_{-1}\hat{S}_z) - \frac{V\beta}{N}(2\Omega_{-1}\hat{S}_x + 2\Delta_{-1}\hat{S}_z)^2, \quad (\text{S31})$$

If we perform a rotation along y-axis with angle $\theta = \arctan \Omega_{-1}/\Delta_{-1}$, then the Hamiltonian become:

$$\begin{aligned} \hat{H}_{\text{eff}} &= -\omega_B \frac{\Delta_{-1}\hat{S}_z - \Omega_{-1}\hat{S}_x}{\sqrt{\Omega_{-1}^2 + \Delta_{-1}^2}} + 2V\sqrt{\Omega_{-1}^2 + \Delta_{-1}^2}\hat{S}_z - 4\frac{V\beta}{N}(\Omega_{-1}^2 + \Delta_{-1}^2)\hat{S}_z^2 \\ &= -4\frac{V\beta}{N}(\Omega_{-1}^2 + \Delta_{-1}^2)\hat{S}_z^2 + \frac{\omega_B\Omega_{-1}}{\sqrt{\Omega_{-1}^2 + \Delta_{-1}^2}}\hat{S}_x - \left(\frac{\omega_B\Delta_{-1}}{\sqrt{\Omega_{-1}^2 + \Delta_{-1}^2}} - 2V_1\sqrt{\Omega_{-1}^2 + \Delta_{-1}^2}\right)\hat{S}_z, \end{aligned} \quad (\text{S32})$$

which takes the form of the LMG model and gives $\tilde{\chi}$, $\tilde{\Omega}$, $\tilde{\delta}$ defined in the main text.

S3. SCHWINGER BOSONS AND UNDEPLETED PUMP APPROXIMATION

In this part, we start with the effective Hamiltonian [Eq. (S14)], but consider the shallow lattice region around $6E_R$. The associated $J_{m,n}$ couplings are plotted in the right panel of Fig. S1. Instead of being localized in a single lattice site, the WS states can extend over a few adjacent lattice sites in a shallow lattice. This can lead to significant suppression of differential AC Stark shifts (homogeneous $J_{n,n}$) at the so-called magic lattice depth ($V_0 = 6E_R$ in our case). Note that the energy difference between nearest-neighbour WS states is rough ω_B , which allows us to study Bloch oscillations under cavity-mediated interaction. We consider the WS states with index $m, n \in \{-1, 0, 1\}$ and use undepleted pump approximation (UPA) $\hat{c}_0 \approx \sqrt{N}$ which is valid at short times when starting from $|\phi_0\rangle$,

$$\hat{N}_{\text{eff}} \approx 2\Delta_1\hat{c}_1^\dagger\hat{c}_1 + 2\Delta_{-1}\hat{c}_{-1}^\dagger\hat{c}_{-1} + \sqrt{N}[\Omega_1(\hat{c}_1^\dagger + \hat{c}_1) - \Omega_{-1}(\hat{c}_{-1}^\dagger + \hat{c}_{-1})] + NJ_{0,0} \quad (\text{S33})$$

$$\equiv \hat{O} + NJ_{0,0}. \quad (\text{S34})$$

Since $\hat{O}$ is small under UPA that assumes the $\hat{c}_{\pm 1}$ modes remain almost unoccupied, we can expand the effective Hamiltonian [Eq. (S14)] up to second order in $\hat{O}$, and ignore the higher-order terms. The term $\hat{O}^2$ we can be approximated to be:

$$\hat{O}^2 \approx N[\Omega_1(\hat{c}_1^\dagger + \hat{c}_1) - \Omega_{-1}(\hat{c}_{-1}^\dagger + \hat{c}_{-1})]^2, \quad (\text{S35})$$

and then the effective Hamiltonian becomes:

$$\hat{H}_{\text{eff}}/\hbar = \omega_B(\hat{c}_1^\dagger \hat{c}_1 - \hat{c}_{-1}^\dagger \hat{c}_{-1}) - \frac{VN/\beta}{1 + \beta J_{0,0} + \beta \hat{O}/N} \quad (\text{S36})$$

$$\approx \omega_B(\hat{c}_1^\dagger \hat{c}_1 - \hat{c}_{-1}^\dagger \hat{c}_{-1}) + V_1 \hat{O} + V_2 \hat{O}^2 \quad (\text{S37})$$

$$= \omega_B(\hat{c}_1^\dagger \hat{c}_1 - \hat{c}_{-1}^\dagger \hat{c}_{-1}) + V_1 \sqrt{N}[\Omega_1(\hat{c}_1^\dagger + \hat{c}_1) - \Omega_{-1}(\hat{c}_{-1}^\dagger + \hat{c}_{-1})] \quad (\text{S38})$$

$$+ V_1(\Delta_1 \hat{c}_1^\dagger \hat{c}_1 + \Delta_{-1} \hat{c}_{-1}^\dagger \hat{c}_{-1}) + V_2 N[\Omega_1(\hat{c}_1^\dagger + \hat{c}_1) - \Omega_{-1}(\hat{c}_{-1}^\dagger + \hat{c}_{-1})]^2, \quad (\text{S39})$$

here

$$V_1 = \frac{2V}{(1 + \beta J_{0,0})^2}, \quad V_2 = -\frac{V\beta/N}{(1 + \beta J_{0,0})^3}. \quad (\text{S40})$$

Moreover, we can absorb the linear term generated by single-particle tunneling via a displacement of a coherent state, $\hat{c}_{\pm 1} = \alpha_{\pm 1} + \hat{c}'_{\pm 1}$ to obtain,

$$\hat{H}_{\text{eff}}/\hbar \approx \omega_B(\hat{c}'_1 \hat{c}'_1 - \hat{c}'_{-1} \hat{c}'_{-1}) + V_1 \Delta(\hat{c}'_1 \hat{c}'_1 + \hat{c}'_{-1} \hat{c}'_{-1}) + V_2 N \Omega^2(\hat{c}'_1 + \hat{c}'_1 - \hat{c}'_{-1} - \hat{c}'_{-1})^2 \quad (\text{S41})$$

here we have made the approximation $\Omega_1 \equiv \Omega \approx -\Omega_{-1}$ as well as $\Delta_1 \equiv \Delta = \Delta_{-1}$. The displacements then become,

$$\alpha_1 = \frac{V_1 \sqrt{N} \Omega (V_1 \Delta - \omega_B)}{(V_1 \Delta)^2 - \omega_B^2 - 8\Omega^2 V_2 N \omega_B}, \quad \alpha_{-1} = \frac{V_1 \sqrt{N} \Omega (V_1 \Delta + \omega_B)}{(V_1 \Delta)^2 - \omega_B^2 - 8\Omega^2 V_2 N \omega_B} \quad (\text{S42})$$

The short-time dynamics [Eq. (S39)] can be calculated analytically for the quadratic Hamiltonian in terms of $\hat{c}'_{\pm 1}, \hat{c}'_{\pm 1}^\dagger$ with $\hat{c}'_{\pm 1} = i[\hat{H}_{\text{eff}}/\hbar, \hat{c}_{\pm 1}]$ and $\hat{c}'_{\pm 1}^\dagger = i[\hat{H}_{\text{eff}}/\hbar, \hat{c}_{\pm 1}^\dagger]$. In this limit the dynamics is given by the equation:

$$i \frac{d}{dt} \begin{pmatrix} \hat{c}'_1 \\ \hat{c}'_{-1} \\ \hat{c}'_1^\dagger \\ \hat{c}'_{-1}^\dagger \end{pmatrix} = \mathcal{H}_{\text{BdG}} \begin{pmatrix} \hat{c}'_1 \\ \hat{c}'_{-1} \\ \hat{c}'_1^\dagger \\ \hat{c}'_{-1}^\dagger \end{pmatrix}, \quad (\text{S43})$$

with the coupling matrix S :

$$\mathcal{H}_{\text{BdG}} = \begin{pmatrix} \omega_B + V_1 \Delta + 2V_2 \Omega^2 & 2V_2 \Omega^2 & -2V_2 \Omega^2 & -2V_2 \Omega^2 \\ -2V_2 \Omega^2 & -2V_2 \Omega^2 & -\omega_B + V_1 \Delta + 2V_2 \Omega^2 & 2V_2 \Omega^2 \\ -2V_2 \Omega^2 & -\omega_B - V_1 \Delta - 2V_2 \Omega^2 & 2V_2 \Omega^2 & 2V_2 \Omega^2 \\ 2V_2 \Omega^2 & 2V_2 \Omega^2 & -2V_2 \Omega^2 & \omega_B - V_1 \Delta - 2V_2 \Omega^2 \end{pmatrix}. \quad (\text{S44})$$

The matrix $\mathcal{H}_{\text{BdG}}$ can have either real or complex eigenvalues, which leads to distinct dynamical behaviors as shown in the main text. When all the eigenvalues are real, the populations $\rho_{\pm 1}$, with $\rho_n = \langle \hat{c}_n^\dagger \hat{c}_n \rangle$, feature stable small amplitude oscillations; on the other hand when all the eigenvalues are complex, then $\rho_{\pm 1}$ feature an exponential growth associated with the correlated pair production of atoms at WS centered at $n = \pm 1$, which leads to the amplification of the Bloch oscillation signal until the UPA breaks down.

S4. EXPERIMENTAL CONSIDERATIONS

A. Single-particle Bloch oscillations

In this section, we discuss the protocols to observe single-particle Bloch oscillations in the experiment. The main idea is to prepare a superposition of different WS states which accumulate different phases under $\hat{H}_0$. In the main

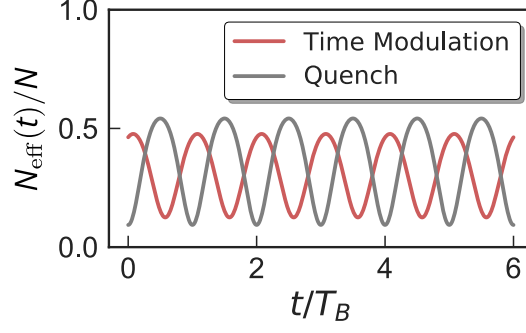


FIG. S4. Single particle Bloch oscillation with amplitude modulation scheme (pink curve, $V_0 = 8E_R$ and $V_1 = 0.4E_R$) and Quench scheme (grey curve, quench from $V_0 = 15E_R$ to $V_0 = 8E_R$ as main text).

text, we discussed the quench scheme where the initial localized WS state ϕ_0 becomes a superposition of delocalized WS states. An alternative way to probe Bloch oscillations is to amplitude modulate the lattice depth as:

$$\begin{aligned} H_1(t)/\hbar &= V_1 \sin^2 k_l z \cos(\omega t + \phi) \\ &= \sum_{m,n=-\infty}^{\infty} t_m \cos(\omega t + \phi) (\hat{c}_{m+n}^\dagger \hat{c}_n + \hat{c}_n^\dagger \hat{c}_{m+n}), \end{aligned} \quad (\text{S45})$$

which has been demonstrated in [S3]. Here we define the tunnelling rate between ϕ_{m+n} and ϕ_n as:

$$t_m = V_1 \int dz \sin^2(k_l z) \phi_{m+n}(z) \phi_n(z), \quad (\text{S46})$$

Moreover, we can choose $\omega \approx m\omega_B, m \in \mathbb{Z}$ to drive the m th sideband (between ϕ_{m+n} and ϕ_n) and ignore the fast rotating terms:

$$\begin{aligned} H_1(t)/\hbar &= \sum_{m,n=-\infty}^{\infty} t_m \cos(\omega t + \phi) (e^{im\omega_B t} \hat{c}_{m+n}^\dagger \hat{c}_n + e^{-im\omega_B t} \hat{c}_n^\dagger \hat{c}_{m+n}) \\ &\approx \sum_{n=-\infty}^{\infty} \frac{t_m}{2} (e^{-i\phi} \hat{c}_{m+n}^\dagger \hat{c}_n + e^{i\phi} \hat{c}_n^\dagger \hat{c}_{m+n}). \end{aligned} \quad (\text{S47})$$

As a result, starting from ϕ_0 and performing the amplitude modulation for time τ , we obtain the initial state to be a superposition of WS states $\{\phi_{n \times m}\}$. In Fig. S4 (pink curve), we simulate the case with lattice depth $V_0 = 8E_R$ and modulation strength $V_1 = 0.4E_R$, also the first sideband transition ($\omega = \omega_B$). Different from the quench scheme (grey curve), after the modulation the single particle wavefunction can have a non-zero coupling to the cavity field ($N_{\text{eff}}/N \neq 0$).

In the experiment, there may be higher bands populated in the quench protocol we discussed in the main text. In other words, the WS basis describing the ground band for the shallow lattice (after quench) is not necessarily complete to describe the initial localized state. Higher bands population will inevitably introduce other frequency components to $N_{\text{eff}}(t)$ disrupting the BO signal. However, in the simulations we performed for the main text (quench from $V_0 = 15E_R$ to $V_0 = 8E_R$), 98% atoms remained in the ground band and the higher band population can be ignored. Similarly, in the amplitude modulation schemes, we also choose V_1 to be much smaller than the band gap to avoid higher bands population.

B. Experimental parameters

Here we discuss the parameters for the specific case of ^{87}Rb with incommensurate lattice wavelength ($\lambda_l = 532 \text{ nm}$, $\omega_B = 2\pi \times 557 \text{ Hz}$) and cavity wavelength (D_2 transition with $\lambda_c = 780 \text{ nm}$). We are interested in the parameter regime with $V \sim \omega_B$ and $\beta \sim O(1)$, where the dynamics is mostly unitary and the dissipative processes can be ignored as we explain below. Another requirement is that the band gap ($27\omega_B$ for $\lambda_l = 532 \text{ nm}$ and $V_0 = 6E_R$) should

Atomic species	λ_l (nm)	λ_c (nm)	Magic lattice depth (E_R)
^{87}Rb (boson)	532	780	$6E_R$
^{87}Sr (fermion)	532	689	$5E_R$
^{88}Sr (boson)	532	689	$5E_R$
^{171}Yb (fermion)	413	556	$3.2E_R$

TABLE S1. Summarized lattice, cavity wavelength and magic lattice depth for different atomic species.

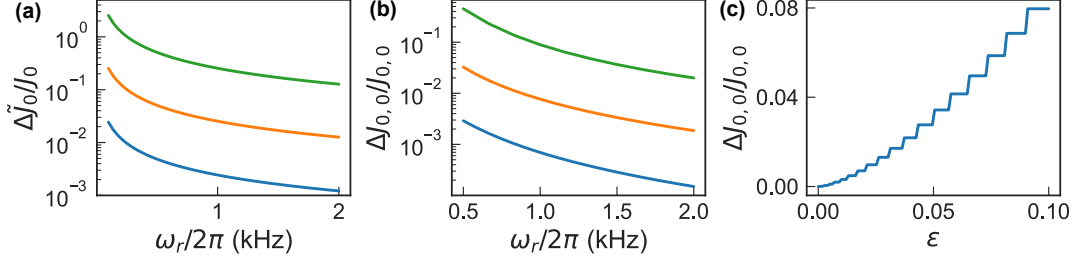


FIG. S5. The standard deviations of (a) the ground band tunneling rates, (b) the coupling coefficient $J_{0,0}$ as a function of radial trapping ω_r with fixed $T = 0.1 \mu\text{K}$ (blue curve), $T = 1 \mu\text{K}$ (orange curve) as well as $T = 10 \mu\text{K}$ (green curve). We use the beam width $w_l = 50 \mu\text{m}$ and lattice potential $V_0 = 6E_R$ in the calculation. (c) The standard deviations of $J_{0,0}$ as a function of loading error rate ϵ .

be much larger than V if we want to only work with the ground band WS states. Moreover, the cavity decay rate $\kappa \sim 2\pi \times 0.1 \text{ MHz}$, the atom-light coupling strength $\mathcal{G}_0 \sim 2\pi \times 0.3 \text{ MHz}$, and atomic transition decay rate $\gamma \sim 2\pi \times 10 \text{ MHz}$ give the cavity cooperativity $C = 4\mathcal{G}_0^2/\gamma\kappa \sim 0.36$, which can be tuned even larger for larger $\mathcal{G}_0$ and smaller κ, γ . The cavity loss generates collective dephasing processes at a rate $V\beta\kappa/\Delta_c$, while spontaneous emission generates off-resonant photon scattering processes at a rate $V\gamma/\Delta_0$ as mentioned in the main text. Under $\kappa/\Delta_c \sim 0.05$ and $\gamma/\Delta_0 \sim 0.01$, one obtains negligible dissipation within the experimentally relevant time scales and $\beta \sim O(1)$. For the maximum AC Stark shift, we first find that $\mathcal{G}_0^2/\Delta_0 \sim 2\pi \times 100 \text{ Hz}$ with the parameters listed above, then $|\eta_p|^2/\Delta_c^2$ can be tuned between 1 to 10 for $V \sim \omega_B$.

Our proposal works with a single internal level in the ground state manifold for atoms hopping between motional states (WS states here). Since interactions are mediated by photons, quantum statistics are not important in our scheme. As a result, even though above we considered the case of Rb, our model can be realized with other species of alkali atoms (D_2 transition) and alkaline earth atoms ($^1S_0 \rightarrow ^3P_1$ transition) i.e. ^{87}Sr (boson), ^{88}Sr (fermion), ^{171}Yb (fermion) with appropriate choices of lattice wavelength and magic lattice depth summarized in table S1. Note that both ^{88}Sr , ^{171}Yb have very small scattering lengths in the ground states.

The single particle Bloch oscillations and dynamical phase transition in the deep lattice doesn't set too much limit on the choice of λ_l and λ_c . We only want the near-neighbour coupling coefficient $J_{m,m+1}$ to be larger, while the overlaps between $\phi_m(z)$, $\phi_{m+1}(z)$ and $\sin^2(k_c z)$ become tiny when $\lambda_l \approx \lambda_c$, so we want to choose different λ_l and λ_c . While for the amplification of BOs in the shallow lattice region, we need to perform the experiment around magic lattice depth thus too shallow magic depth (such as ^{171}Yb) isn't favorable.

C. Radial mode thermal distribution

In this section, we discuss the effect of the Radial thermal motion following Ref. [S4]. The Gaussian geometry of the laser beams in experiments inevitably couples the vertical and radial wave functions. The Gaussian profile of the lattice and cavity beams causes atoms in different radial modes to have different tunneling rate, resulting in a slightly different overlap integral $J_{m,n}$ for atoms in different radial modes. This effect can also be understood as fluctuations of the lattice potential V_0 due to radial thermal excitation.

First, we focus on the Gaussian beam profile of a 1D lattice, which leads to the following trapping potential:

$$V_0(r, z) = V_0 \sin^2(k_l z) \exp(-2r^2/w_l^2), \quad (\text{S48})$$

where w_l is the beam width. In the presence of additional radial trapping potential $V_r(r, z) = M\omega_r^2 r^2/2$ [S5] we can

expand the total trapping potential $V(r, z) = V_0(r, z) + V_r(r, z)$ to second order of r and obtain:

$$V(r, z) \approx V_0 \sin^2(k_l z) + \frac{1}{2} M \omega_r^2 r^2 - \frac{\omega_{r0}^2}{\omega_r^2} \frac{1}{2} M \omega_r^2 r^2 \sin^2(k_l z), \quad (\text{S49})$$

here $\omega_{r0} = \sqrt{4V_0/M\omega_l^2}$.

The first term describes the lattice potential along the axial direction with the characteristic Bloch functions as eigenstates. The second term describes the radial harmonic trapping with eigenstate $\phi_{n_x, n_y}(r) = \phi_{n_x}(x)\phi_{n_y}(y)$ and eigenenergies $E_{n_x, n_y} = \hbar\omega_r(n_x + n_y + 1/2)$. The third term describes the coupling between axial and radial degrees of freedom. The correction of J_0 is given by [S4]:

$$\tilde{J}_0(n_x, n_y) = J_0 + \frac{1}{8} \frac{\omega_{r0}^2}{\omega_r^2} E_{n_x, n_y} \left[\frac{\partial}{\partial v_0} f(\tilde{q} = 0, v_0/4) - \frac{\partial}{\partial v_0} f(\tilde{q} = \pm 1, v_0/4) \right]. \quad (\text{S50})$$

Here the function f is the characteristic Mathieu value of type A for $q \in (-\hbar k_l, \hbar k_l)$, and the characteristic Mathieu value of type B for $q = \pm k_l$. We define $\tilde{q} = q/\hbar k_l$ and $v_0 = V_0/E_R$. One can take such $\tilde{J}_0$ into Eq. (S7) to calculate $\phi_n(z)$, which causes inhomogeneity for the coupling matrix $J_{m,n}$. We can estimate the contribution from different radial eigenmode with Boltzmann distribution $p_{n_x, n_y} = \exp[-(n_x + n_y)\omega_r\hbar/k_B T]/Z$, in which the partition function $Z \approx (k_B T/\hbar\omega_r)^2$. Then we can calculate the variance of the $\tilde{J}_0$ as:

$$\begin{aligned} \Delta \tilde{J}_0^2 &= \sum_{n_x, n_y} p_{n_x, n_y} [\tilde{J}_0(n_x, n_y) - J_0]^2 \\ &= \left\{ \frac{\hbar \omega_{r0}^2}{8 \omega_r} \left[\frac{\partial}{\partial v_0} f(\tilde{q} = 0, v_0/4) - \frac{\partial}{\partial v_0} f(\tilde{q} = \pm 1, v_0/4) \right] \right\}^2 \frac{2(e^{\hbar\omega_r/k_B T} + 2)}{(e^{\hbar\omega_r/k_B T} - 1)^2}. \end{aligned} \quad (\text{S51})$$

In Fig. S5(a), we plot the standard deviation of the tunneling rate J_0 as a function of ω_r and different temperature T . In Fig. S5(b), we plot the standard deviation of the coupling coefficient $J_{0,0}$ due to the correction of the tunneling rate. Similar behavior for other coupling coefficients $J_{m,n}$. As a result, one can suppress the effect of the radial modes occupation by increasing the total radial trapping frequency ω_r or lowering the temperature. The standard deviation $\Delta J_{0,0} \approx 0.01 J_{0,0}$ up to temperature $T \sim 1 \mu\text{K}$ as well as $\omega_r = 2\pi \times 1 \text{ kHz}$, thus the radial thermal noise only has a tiny effect on many-body dynamics we predict.

D. Atoms loading

In this section, we discuss the real process of atoms loading in the experiment. In the main text, we mention first loading atoms at position $k_c z/\pi = r, r \in \mathbb{Z}$ which atoms-cavity coupling becomes perfect zero. However one can only set a threshold for the atom-cavity coupling during the loading process i.e. load all the atoms with $\sin^2 k_c z < \epsilon$ in the real experiment. Such loading error makes $J_{m,n}$ deviate from expected values, which brings additional inhomogeneity. In Fig. S5(c), we plot the standard deviation of the coupling coefficient $J_{0,0}$ as a function of error ϵ . We consider the total lattice length to be 1 mm and assume atoms load into all the sites n which satisfy $\sin^2(k_c n a_l) < \epsilon$ uniformly. These imperfect sites cause tiny inhomogeneity in the coupling coefficient $J_{m,n}$ up to $\epsilon \sim 5\%$.

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